

PEDDA REDDAPPA KANAKANTI
Lead AI & Data Platform Engineer

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Portfolio: kporreddy.github.io/portfolio-site/

PROFESSIONAL SUMMARY

Lead AI and Data Platform Engineer / Architect with 15+ years of experience building and operating secure, high-scale platforms across AWS, Databricks, Snowflake, Spark, Kafka, and open lakehouse ecosystems. Lead real-time data engineering, cloud modernization, production ML, and governed AI agent and MCP solutions from architecture through production, with accountability for security, reliability, performance, observability, and cost.

- **Enterprise AI and data platforms** — architecture and hands-on delivery spanning ingestion, processing, open lakehouse storage, analytics, production ML, governed AI agents, MCP, security, and operations.
- **Databricks platform architecture on AWS** — multi-account landing zones, workspace strategy, customer-managed VPCs, PrivateLink, Transit Gateway, IAM, KMS, S3, Unity Catalog, SSO/SCIM, and Terraform.
- **Real-time open lakehouse at scale** — actual production volume of approximately 21.4 billion game and AdTech events per day, 6x the earlier baseline, with Kafka buffering and backpressure controls for higher volumes and analytics-ready data available in approximately 20 minutes.
- **AI agents, MCP, and production ML** — governed analytics assistants, semantic layers, RAG, vector search, MLflow, and a productized Kubernetes inference platform with actual production throughput of approximately 6,000 requests per second sustained for more than three months.
- **Technical leadership and measurable impact** — realized approximately \$2.8M in annual savings, improved analytics performance by 5–7x while supporting 4x data volume, reduced warehouse spend by approximately 30%, and eliminated a recurring nightly restart cycle.

CORE TECHNICAL SKILLS

Cloud & Big Data	AWS (S3, Glue, Lambda, EMR, Redshift, Kinesis, DMS, IAM, CloudWatch, CloudTrail, VPC, PrivateLink, Transit Gateway, VPC Endpoints, KMS, Organizations), Databricks, Snowflake
Languages & Frameworks	Python (FastAPI, Flask), PySpark, Scala, SQL
Lakehouse & Storage	Delta Lake, Apache Iceberg, Parquet, Unity Catalog, Medallion Architecture (Bronze/Silver/Gold)
Orchestration & Transformation	Airflow, dbt, Step Functions
Streaming	Kafka, Kinesis, Spark Structured Streaming, Apache Flink
DevOps & Governance	Terraform (IaC), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), IAM, SSO/SAML, SCIM, Databricks Cluster Policies, GDPR/CCPA/PCI-DSS
Databases	Redshift, RDS, DynamoDB, Cassandra
AI & ML Platforms	AI Agents, MCP Hosts and Servers, RAG, Vector Search, MLflow, Model Serving, Feature Stores, LLMops, Snowflake Cortex AI

PROFESSIONAL EXPERIENCE

Senior Data Lead / Architect

Sep 2023 – Present

MobilityWare — Plano, TX

Lead the architecture and hands-on engineering of enterprise data and AI platforms across real-time event processing, open lakehouse, analytics, ML infrastructure, and governed AI integrations, with end-to-end ownership from design through production delivery.

- **Lead AI and data platform architecture** — own technical direction, design reviews, delivery planning, and production operations across AWS, Databricks, Snowflake, Spark, Kafka, and open lakehouse workloads.
- Designed and delivered a production real-time open lakehouse with actual production volume of approximately 21.4 billion game and AdTech events per day, 6x the earlier 3.573 billion-event baseline; implemented Kafka buffering, backpressure controls, replayable offsets, Spark/EMR processing, Apache Iceberg on Amazon S3, schema enforcement, and UUID deduplication to absorb higher event volumes while providing analytics-ready data in approximately 20 minutes.
- Architected a secure Databricks platform on AWS covering multi-environment workspace strategy, customer-managed VPCs, subnet and CIDR planning, PrivateLink, Transit Gateway, S3 endpoints, IAM, KMS, Unity Catalog, SSO/SAML, SCIM, cluster policies, and audit logging.
- Designed and implemented the Snowflake account and environment strategy, secure Amazon S3 integrations, external Iceberg tables, workload-isolated warehouses, role-based access controls, and performance and cost-governance standards; reduced warehouse spend by approximately 30%.
- Architected governed AI agent and MCP integrations that translate business questions into controlled, read-only data access using approved semantic definitions, role-based authorization, query limits, and end-to-end auditability.
- Lead a productized ML inference platform sustaining approximately 6,000 requests per second for more than three months. During production validation, observed 41 ms p95 and 98 ms p99 request-to-model latency; established Kubernetes autoscaling, progressive rollout, observability, recovery, and rollback controls.

Senior Data Lead

Dec 2022 – Sep 2023

Masscom Corporation — Bengaluru, India

Acted as the senior technical contributor and de facto team lead for the AdTech data engineering pod, designing reusable ingestion frameworks and mentoring junior engineers on Databricks, PySpark, and AWS Glue integration patterns.

- Led architecture and hands-on development for AdTech data platforms using Databricks, PySpark, Kafka, AWS Glue, S3, Snowflake, dbt, and Airflow across streaming and batch workloads.
- Provisioned Databricks workspaces on AWS with Terraform, private subnets, VPC endpoints, IAM roles, instance profiles, secure S3 integration, cluster access modes, and environment-specific controls.
- Built reusable ingestion frameworks supporting schema evolution, semi-structured data, incremental processing, and near-real-time delivery into lakehouse and Snowflake analytics layers.
- Established Medallion and open lakehouse patterns using Delta Lake and Apache Iceberg, with centralized governance through Unity Catalog and AWS IAM.
- Optimized clusters through instance-family selection, autoscaling, instance pools, spot/on-demand strategies, runtime policies, and cost-allocation tags while protecting workload SLAs.
- Migrated external processing to in-house Kinesis, S3, Spark, and Python pipelines, reducing operating costs by approximately 50%.
- Improved reliability through Airflow orchestration, metadata audits, data-quality controls, retry and alerting standards, CloudWatch monitoring, and documented platform operating procedures.

Tech Lead

Apr 2020 – Dec 2022

Legato Health Technologies — Bengaluru, India

Served as **Tech Lead** for the healthcare data engineering group, owning technical direction for cloud migration initiatives, leading a cross-functional engineering team, and driving multi-million dollar cost-saving migrations from legacy on-prem systems to AWS and Snowflake.

- Led the healthcare data engineering team and technical direction for migration of regulated Teradata and Oracle workloads to AWS and Snowflake.
- Designed the AWS landing zone, VPC and subnet architecture, Transit Gateway connectivity, VPC endpoints, IAM roles, KMS encryption, S3 storage controls, and audit foundations required for secure cloud adoption.

- Introduced Terraform-based infrastructure provisioning and reviewed CI/CD promotion controls across development, test, and production environments.
- Directed Spark, Scala, EMR, Glue, Lambda, Snowflake, and dbt implementation for scalable batch and analytical workloads, including data validation and cutover planning.
- Delivered cloud migration and platform modernization that realized approximately \$2.8M in annual savings through workload optimization and warehouse cost governance.
- Improved platform security and compliance through least-privilege IAM, encryption, CloudTrail, CloudWatch, recovery planning, and governance controls aligned with healthcare requirements.
- Mentored engineers, established Python and Spark engineering standards, and represented the team in architecture, security, and production-readiness reviews.

Data Engineer

Aug 2018 – Mar 2020

Tiger Analytics — Chennai, India

- Built data pipelines using Spark/Scala and Kafka/Cassandra for real-time and batch processing; leveraged HDFS, Sqoop, Hive, and Pig for large-scale data management.
- Used AWS EMR, Lambda, Redshift, Glue, EC2, and S3 to build scalable ETL pipelines; implemented CI/CD with Jenkins and GitHub for Python data pipelines and REST API services.
- Administered Hadoop clusters and contributed to the migration from Hortonworks (HDP) to Cloudera (CDP).

Hadoop Developer

Jul 2015 – May 2018

Accenture Solutions — Hyderabad, India

- Developed data pipelines using Sqoop, Pig, and Spark; designed Hive tables and HQL queries for comprehensive reporting and data analysis.

Hadoop Developer

Nov 2014 – Jul 2015

Tesco HSC — Bengaluru, India

- Maintained data quality, metadata, and lineage for retail sales, inventory, and customer datasets; implemented automated data validation checks in production pipelines.

Mainframe Developer

Aug 2011 – Oct 2014

UST Global — Bengaluru, India

- Mentored team members, monitored job execution, and led post-implementation reviews, testing, and support.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate, 08/2021
- AWS Certified Database – Specialty, 10/2022
- Snowflake – SnowPro Core Certification, 05/2025
- Databricks – Databricks Data Engineer Associate, 05/2025
- Databricks – Databricks Data Engineer Professional, 06/2025

EDUCATION

Bachelor's Degree, Electronics and Communication Engineering — Jawaharlal Nehru University, Anantapur (05/2005 – 05/2009)